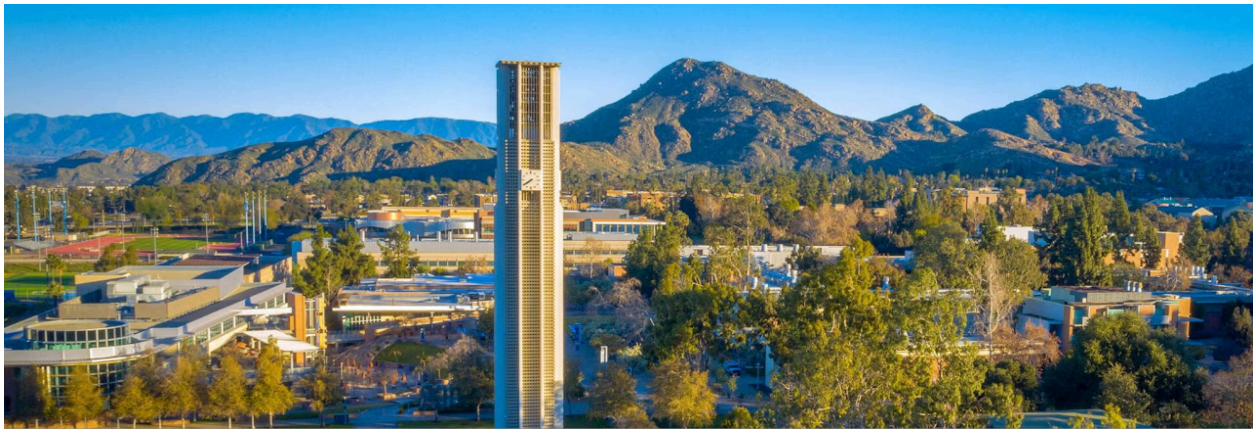
42nd Jim Isenberg Pacific Coast Gravity Meeting

UCR Center for Experimental Cosmology & Instrumentation

The 42nd Jim Isenberg Pacific Coast Gravity Meeting will be held on **Friday, February 27, 2026 - Saturday, February 28, 2026** at the University of California, Riverside.



Program

Thursday, February 26

5:00 PM - 6:00 PM **Welcome Reception**
University Lecture Hall (map)

6:00 PM - 7:00 PM **Frontiers of Cosmology Public Lecture**
University Lecture Hall (map)

Featuring APS President Brad Marston, with introductory remarks by 2017 Nobel Laureate Barry Barish

See the event page for more information.

Friday, February 27

8:30 AM - 9:00 AM **Coffee & Continental Breakfast**
HUB 302 North (map)

9:00 AM - 10:30 AM **Session I: Gravitational-Wave Data Science**
HUB 302 North (map)
Session Chair: Jonathan Richardson

- Barry Barish, University of California, Riverside
  Welcome and Overview
- Ryan Johnson, California State University, Fullerton
  Including Gravitational Waveform Uncertainty in Bayesian Inference of Source Properties
- Joseph Ribaudo, Embry-Riddle Aeronautical University
  Semi-Classical Model of Detector Efficiency in the Context of Bipartite Gravitational-Wave Detection
- Sophie Bini, California Institute of Technology
  The impact of waveform systematics and Gaussian noise on the interpretation of GW231123

- Tousif Islam, University of California, Santa Barbara
  Binary black hole recoil kicks: modeling and astrophysical implications

- Anjana Ashok, Oregon State University
  Long-lived gravitational wave signals in current experiments

- Siddharth Soni, University of California, Riverside
  Machine Learning and Automation in LIGO Detector Characterization

10:30 AM - 11:00 AM **Coffee Break**

11:00 AM - 12:20 PM **Session II: Numerical Relativity**
HUB 302 North (map)
Session Chair: Jocelyn Read

- Andrew Laeuger, California Institute of Technology
  Probing Early Ringdown with the Direct Wave

- Iago Mendes, California Institute of Technology
  Numerical relativity for LISA sources: binary black hole initial data

- Isabella Pretto, California Institute of Technology
  Invariant identification of light rings in binary black hole simulations

- Sepehr Salamat, University of Arizona
  Black Hole Meissner Effect and Singular Extremal Horizons

- Vittoria Tommasini, California Institute of Technology
  Data-Driven Acceleration of Eccentricity Reduction for Binary Black Hole Simulations

12:20 PM - 1:50 PM **Lunch Break**

1:50 PM - 3:20 PM **Session III: Gravity Theory**
HUB 302 North (map)
Session Chair: Preston Jones

- Logan Finke, Embry-Riddle Aeronautical University
  Signatures of non-classicality in gravitational waves through bipartite detections

- Quentin Bailey, Embry-Riddle Aeronautical University
  Tests of spacetime symmetries using an effective field theory framework

- Sawyer Star, Embry-Riddle Aeronautical University
  Properties of Lorentz-symmetry breaking with a modified scalar field

- Arunima Das, Arizona State University
  Coherent and Squeezed States in Gravity

- Colin Weller, California Institute of Technology
  Isospectrality and Chiral Symmetry

- Giuseppe Meluccio, Scuola Superiore Meridionale
  Pre-geometric Gravity and Emergent Cosmology

- Sam Gralla, University of Arizona
  Can black holes evaporate past extremality?

3:20 PM - 3:50 PM **Coffee Break**

3:50 PM - 5:30 PM **Session IV: Gravitational-Wave Detector Science**
HUB 302 North (map)
Session Chair: Joshua Smith

- Tyler Rosauer, University of California, Riverside
  Next-Generation Wavefront Control Technology for Gravitational-Wave Detectors

- Pooyan Goodarzi, University of California, Riverside
  Impact of Beamsplitter Thermal Distortions on the Sensitivity of High-Power Gravitational-Wave Detectors

- Kielo Hoops, California State University, Los Angeles
  Characterization of Optical Scatterers in Coating Materials

- Alina Sendowsky, California State University, Fullerton
  Optical Scatter and Non-Contact Temperature Sensing of Cryogenically Cooled Crystalline Silicon

- Maryam Ranjbar, University of California, Riverside
  Automated Beam Repositioning for Mitigation of Nonuniform Absorption in Gravitational-Wave Detectors

- Cynthia Liang, University of California, Riverside
  Mitigating Point Absorber Risk in Gravitational-Wave Detectors

- Michael Padilla, University of California, Riverside
  Toroidal Mirrors for Enhanced Sensitivity in Current and Next-Generation Gravitational-Wave Detectors

- Xuesi Ma, University of California, Riverside
  Modeling and Characterizing Scattered Light Noise in the LIGO Detectors

5:30 PM - 7:30 PM **Networking Reception**
The Stable (map)
Light refreshments will be served, with drinks available for purchase.

Saturday, February 28

8:30 AM - 9:00 AM **Coffee & Continental Breakfast**
HUB 302 North (map)

9:00 AM - 10:30 AM **Session V: Gravity Theory**
HUB 302 North (map)
Session Chair: Hai-Bo Yu

- Dario Walter-Cardona, Embry-Riddle Aeronautical University
  Systems of PDEs in Bumblebee Gravity

- Ziyi Li, University of California, Davis
  Aspects of Overspinning and Extremal Defects in AdS_3

- Anajak Phonchantuek, California State University, Fresno
  Nonlinear chiral forms in Sen formulation of superstring theory

- Eduardo Guendelman, Ben-Gurion University
  Dark Energy and Dark Matter with Dynamical Tension Strings

10:30 AM - 11:00 AM **Coffee Break**

11:00 AM - 12:20 PM **Session VI: Numerical Relativity**
HUB 302 North (map)
Session Chair: Geoffrey Lovelace

- Dongze Sun, California Institute of Technology
  Parameter matching between PN and NR via an inner-zone matching for binary black holes

- Guido Da Re, California Institute of Technology
  Initial data construction for Cauchy-characteristic evolution

- Alexandra Macedo, California State University, Fullerton
  Writing Tensor Equations in SpECTRE

- Jack Kutcka, California State University, Fullerton
  Testing SpECTRE's Outer Boundary Condition and Stability

- Taylor Knapp, California Institute of Technology
  Full control of the parameters of eccentric and precessing SpEC waveforms

- Jose Pineda, California State University, Fullerton
  Comparing SpEC and SpECTRE evolutions of the same binary-black-hole initial data

12:20 PM - 1:50 PM **Lunch Break**

1:50 PM - 3:20 PM **Session VII: Future Experiments**
HUB 302 North (map)
Session Chair: Siddharth Soni

- Amber Romero, California State University, Fullerton
  Improving techniques for identifying and evaluating sites for Cosmic Explorer

- Cecilia Ochoa, University of California, Riverside
  Mirror Design for Optimal Higher Order Mode Placement in Cosmic Explorer

- Briana Chen, California Institute of Technology
  Using quantum metrology to advance fundamental physics

- Mark Nguyen, Northwestern University
  Precision Tests of Gravity with Optically Levitated Nanoparticles

- Sujoy Modak, California State Polytechnic University, Pomona
  A Pathway for Probing Weak Quantum-Gravity Deformations Using Present-Day Technology

- Spencer Johnson, Jet Propulsion Laboratory
  Toward Gravitational Redshift Measurements with Quantum States of Light

- Douglas Singleton, California State University, Fresno
  Temporal Pound-Rebka experiment as gravitational Aharonov-Bohm effect

3:20 PM - 3:50 PM **Coffee Break**

3:50 PM - 5:20 PM **Session VIII: Gravity Theory**
HUB 302 North (map)
Session Chair: Douglas Singleton

- Daniel Hooker, California State University, Fresno
  Symmetrized operators versus modified integration measure in Generalized Uncertainty Principle models

- Zach Ghosoph, Clovis Community College
  From Linearized GR to Quantum Gravity via Canonical Quantization

- Diego Hidalgo, Clovis Community College
  From Quantum Gravity to Linearized General Relativity via Bohr Correspondence Principle

- Rose Zanoni, Clovis Community College
  Resolving the Cosmological Constant Problem via quantum space-time uncertainty

- Nathan Inan, California State University, Fresno
  Evolution of Quantum Space-Time Uncertainty on Cosmological Scales

- Michael Bishop, California State University, Fresno
  Minimal lengths in 3D via the Generalized Uncertainty Principle

5:20 PM - 5:30 PM **Announcement of DGRAV Student Prize Winner**
HUB 302 North (map)